



江苏省苏州中学 国际部
Suzhou High School - International Division

SZZX-ID Bridge Program Physics

Graphs for motion

Class Outline

Title of Unit	Linear motion
Title of Lesson	Graphs for motion
Starter Activity	vocabulary
Direct Instructions	Teaching Strategies: Demonstration, Lecture, Collaboration, Questioning, Inquiry
	1. Lecture: s-t graph and v-t graph 2. Question: Interpret negative slope for the v-t and s-t diagrams 3. Inquiry: compare the graphs
	Differentiation: Content, Instructional Strategies, the Classroom, Products, Teacher
	1. Content: check frequently for previous knowledge on slope of a curve
	International Mindedness:
Lesson Objectives	By the end of the lesson, students will be able to: 1. Understand and interpret s-t and v-t graph 2. Understand the slope of a s-t graph represents v; the slope of a v-t motion represents a 3. Understand the area bounded by a v-t graph is displacement 4. Derive suvat equations from graph
ATLs	Identify unit ATLs and how they are being addressed in the lesson
	communication: Understand Command terms for DP physics
IB Learner Profile	Link the lesson to the characteristics of the IB Learner attribute
	Communicators: represent physic quantity with vectors, interpret graphs
Teaching Activities	1. Activity: interactive v-t curve 2. Worksheet: Compare the motion graphs 3. (Kahoot) graphs for motion
Plenary Session	Checking for understanding / Homework / Q & A / Reflection etc.
	Check feedback from students for the first topic: too much? Too little? Language?

Vocabulary



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- Diagram 图
 - Graph 图表
 - Tangent 切线
 - Secant 割线
 - Gradient 斜率
 - Slope 斜率/斜坡
-
- Steep 陡
 - Flat 平
 - Equation 方程
 - Derive/derivation 推导

Describe motion



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- We can describe motion in words
- We can describe motion with data. e.g. in a table like this

Time (s)	0	100	200	300	400	500	600	700	800
Position (m)	0	200	400	400	400	900	1400	1400	1400

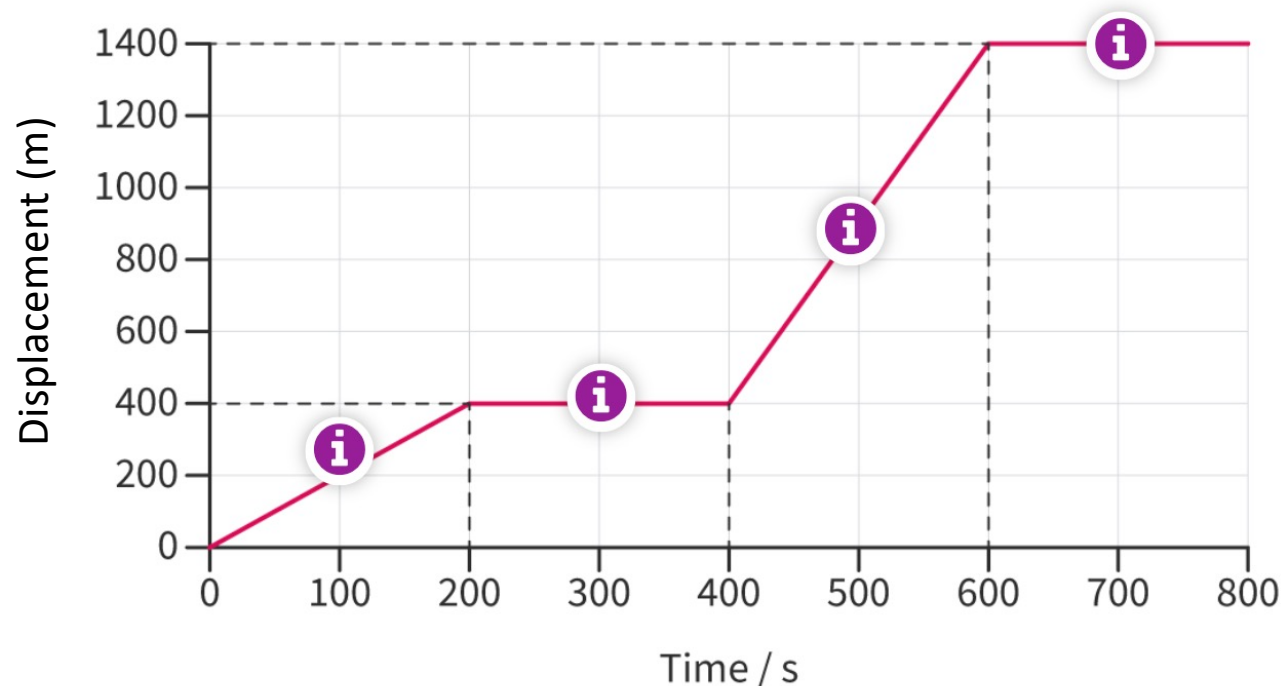
- From the data table above, what can you tell? What can't you tell?

Describe motion: s-t diagram



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- Alternatively, graphs can be used to describe motion
- Displacement vs. time can be plotted



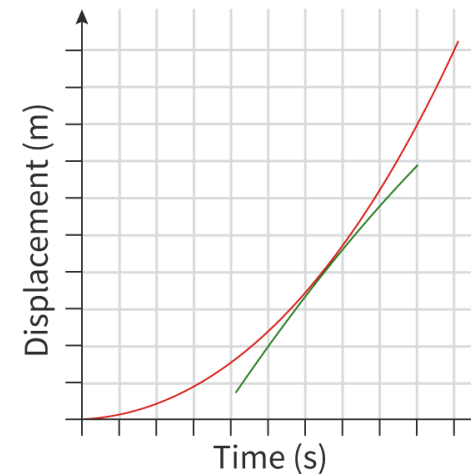
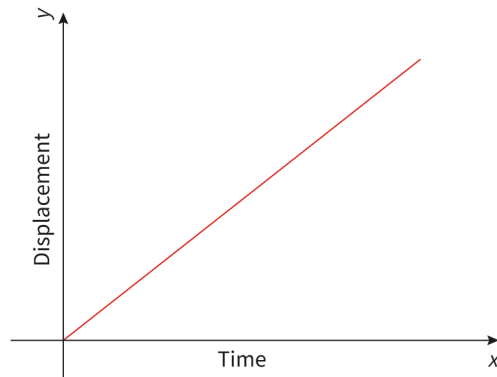
Can you describe the motion in each section on this graph?

Describe motion: s-t diagram



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- The gradient of a displacement–time graph is the velocity.
- The steeper the line, the larger the gradient, and the faster the body is moving
- If the gradient of the graph is not constant, it means the velocity is not constant
- To find the instantaneous velocity of the object at a particular moment in time, you need to draw a tangent to the curve as shown, then calculate the gradient of the line.

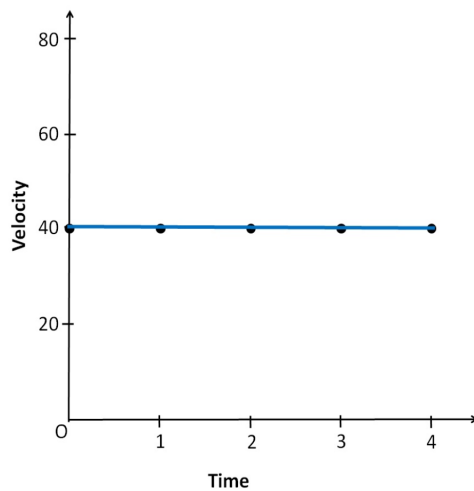


Describe motion: v-t diagram

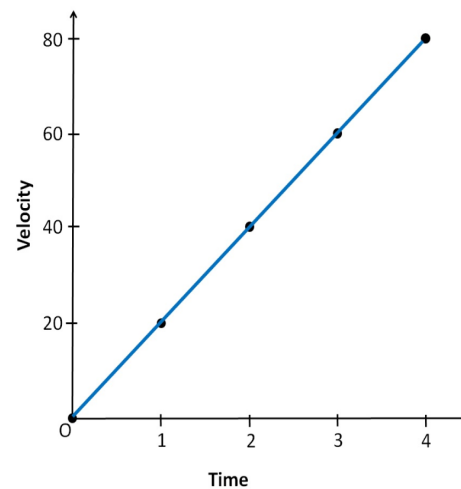


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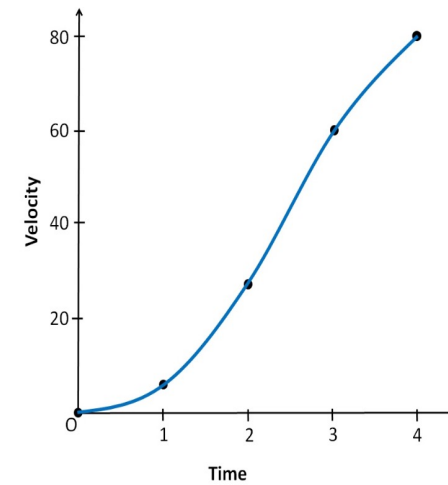
- The gradient of a v-t graph shows acceleration
- The area bounded by a v-t graph shows displacement



Uniform motion
Velocity is constant



Uniformly accelerated motion
Acceleration is constant



Non-uniformly accelerated motion
acceleration is not constant

Equations for motion



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- The table shows quantities that we can use to describe a UAM
- If we know some quantities of a UAM, how do we work out the other quantities?

Quantity	Symbol	Unit
Displacement	s	m
Initial velocity	u	m s^{-1}
Final velocity	v	m s^{-1}
Acceleration	a	m s^{-2}
Time	t	s

Starting from the definition of acceleration



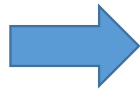
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- Acceleration is the rate of change of velocity

$$a = \frac{\Delta v}{\Delta t}$$

- For uniformly accelerated motion:

$$a = \frac{v-u}{t}$$



$$v = u + at$$

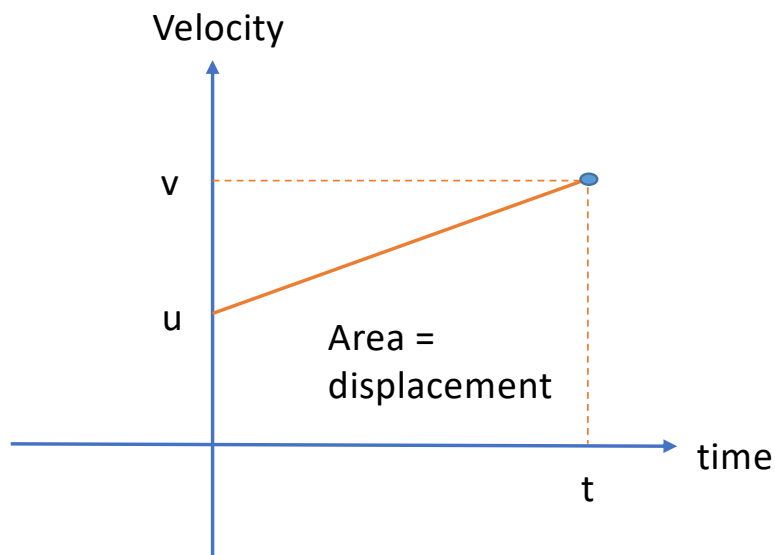
1

Where v and u are the final and initial velocity, t is the interval of the motion

Suvat equations – derivation



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$$S = \frac{(u+v) \times t}{2}$$

2

S: displacement

In ②, substitute v with $v = u + at$

In ②, substitute t with $t = \frac{v-u}{a}$

$$S = \frac{(u+u+at) \times t}{2}$$

$$= ut + \frac{1}{2} at^2$$

3

$$S = \frac{(v+u)(v-u)}{2a}$$

So:

$$v^2 - u^2 = 2as$$

4

Conditions for suvat equations to be applied



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The equations of motion
ONLY APPLY when:

- The motion is in a straight line
- There is constant acceleration

Objects in free fall follow these conditions, when air resistance is ignored.

1

$$v = u + at$$

2

$$s = \left(\frac{v + u}{2} \right) t$$

3

$$s = ut + \frac{1}{2}at^2$$

4

$$v^2 = u^2 + 2as$$